# C2 Problems: exponential and logarithmic functions

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"Bereken exact" means no decimal approximations. Where an approximation is asked, you may use a calculator, but write down the exact expression you are evaluating.

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## Practice

**1.** Compute exactly.

a. $^{2}\log 64$ b. $^{5}\log \frac{1}{125}$ c. $^{8}\log 4$ d. $^{1/3}\log 27$ e. $\ln\left(e^{2}\sqrt{e}\right)$ f. $\log 0.0001$

**2.** Simplify.

a. $^{2}\log 12 + {}^{2}\log 6 - 2 \cdot {}^{2}\log 3$

b. $3\ln x - \ln(x^{2}) + \ln\left(\dfrac{4}{x}\right)$ for $x > 0$

**3.** Solve exactly.

a. $3^{x+1} = 81$ b. $9^{x} = 27^{x-2}$ c. $7^{x} = 20$ (also give a two-decimal approximation) d. $2^{x} \cdot 5^{x} = 0.01$

**4.** Solve exactly.

a. $^{3}\log(2x - 1) = 2$

b. $^{2}\log x + {}^{2}\log(x + 2) = 3$

**5.** Solve exactly: $9^{x} - 4 \cdot 3^{x} + 3 = 0$.

**6.** Caesium-137 has a half-life of 30 years.

a. Find the growth factor per year (four decimals).
b. How long does it take until 5% of a quantity is left? Round to whole years.

**7.** A savings account pays 6% interest per year.

a. Compute the doubling time exactly and rounded to two decimals.
b. Compare with the "rule of 70" from the notes.

**8.** Given $f(x) = 3 \cdot 2^{x+1} - 12$.

a. Give the horizontal asymptote and the range.
b. Bereken exact the zero of $f$.
c. Find $f^{-1}(x)$.

**9.** Solve exactly.

a. $e^{3x} = 7$

b. $\ln x + \ln(x - 2) = \ln 8$

**10.** Lemon juice has pH 2.5; rain water has pH 6.0. How many times as large is the hydrogen-ion concentration in lemon juice? Give an exact expression and an approximation.

**11.** Solve.

a. $0.6^{x} > 0.2$ (give the boundary exactly and to two decimals)

b. $^{1/3}\log(x + 2) < -1$ (exactly)

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## CE-level

**12.** Given $f(x) = \ln(9 - x^2)$.

a. Give the domain of $f$.
b. Solve exactly: $f(x) = \ln 5$.
c. Solve exactly: $f(x) \geq \ln 8$.
d. Toon aan that the maximum value of $f$ is $2\ln 3$.

**13.** Given $f(x) = \ln x$ and $g(x) = \ln(4 - x)$.

a. Toon aan that the graph of $g$ is the reflection of the graph of $f$ in the line $x = 2$.
b. Bereken exact the coordinates of the intersection point of the graphs.
c. Solve exactly: $f(x) + g(x) = \ln 3$.

**14.** After an injection, the concentration of a medicine in the blood is $C(t) = 12 \cdot 0.85^t$ mg/L, $t$ in hours.

a. Calculate the half-life in hours, rounded to two decimals.
b. Write $C$ in the form $C(t) = 12 \cdot e^{-kt}$. Give $k$ exactly and to four decimals.
c. The medicine works as long as $C > 1$. Calculate how long it works, in hours and minutes.

**15.**

a. Prove that ${}^{a}\log b \cdot {}^{b}\log c = {}^{a}\log c$ for positive $a, b, c$ with $a, b \neq 1$.
b. Compute exactly: ${}^{2}\log 3 \cdot {}^{3}\log 4 \cdot {}^{4}\log 5 \cdots {}^{15}\log 16$.

**16.** Given $f(x) = e^{2x} - 4e^x$.

a. Bereken exact the zero of $f$.
b. Solve exactly: $f(x) = 5$.
c. Solve exactly: $f(x) < 0$.

**17.** The sound level of one machine is 80 dB. Sound intensities of machines add up. Use $L = 10 \cdot \log \dfrac{I}{I_0}$.

a. Calculate the sound level of 2 identical machines, to one decimal.
b. How many identical machines give 100 dB?

**18.** The energy $E$ released by an earthquake is related to the magnitude $M$ by $M = \frac{2}{3} \log \dfrac{E}{E_0}$.

a. Toon aan that when $M$ increases by 1, the energy is multiplied by $10^{1.5}$.
b. How many times as much energy does a magnitude 7.2 earthquake release as one of magnitude 5.8?

**19.** In a lab experiment, $\log N$ is plotted against $t$ (in hours). The points lie on the line through $(0,\ 1.5)$ and $(5,\ 4)$.

a. Find the formula $N = b \cdot g^t$. Give $b$ and $g$ exactly and to two decimals.
b. Calculate the doubling time in minutes.

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## Challenge

**20.** Prove that ${}^{2}\log 3$ is irrational.

**21.** Which is larger, $31^{11}$ or $17^{14}$? Decide without a calculator.

**22.** Solve: $x^{\log x} = 100x$ (base-10 logarithm).

**23.** Consider the equation $2^x = x^2$.

a. Find two positive integer solutions.
b. Show that there is also a negative solution, and find it to two decimals with a calculator.
c. (Harder) Explain why there are no further solutions. You may use a graph, but give reasons why the graph can be trusted.

**24.** Solve exactly: ${}^{x}\log 2 + {}^{2}\log x = \frac{5}{2}$.

**25.** Carbon-14 has a half-life of 5730 years. A piece of wood contains 30% of the carbon-14 that living wood contains. Estimate its age, rounded to tens of years.